Mechanistic Interpretability Lab @ Lectures on Computational Linguistics

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What is mechanistic interpretability?

Definition?

- No clear and uniformly accepted definition

- A possible definition that has been often used is (not cited verbatim):

  “Mechanistic interpretability is the attempt to reverse engineering the algorithms implemented by neural networks into human-understandable mechanisms”

For more definitions and a recent history of the field I highly recommend you to read this paper:
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What tools?

Methods used in papers referring to the term mechanistic interpretability

Input attribution:
- Saliency maps
- Integrated Gradients
- LIME
- SHAP

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Weight attribution:
- Low-rank updates
- Pruning / ablation
- Weight decompositions

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Transformer-specific interpretability:
- Attention patterns
- Vocabulary projections
- Circuit analysis
- Induction heads

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Activation-space analysis:
- Linear probes
- **Representation engineering**
- **Sparse Autoencoder**
- Transcoders

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The problem: Interpretable control of LLMs

Concepts

Italian language

Being a pirate

Python

Generated text

“Oggi è una bella giornata!”

“Arrr, matey! The seas be rough today!”

“numbers = [12, 18, 25, 31]
average = sum(numbers) / len(numbers)”

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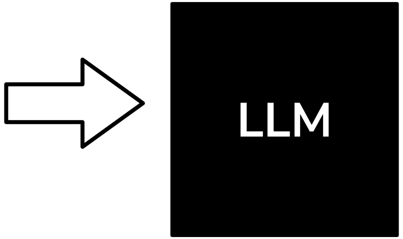
Prompting

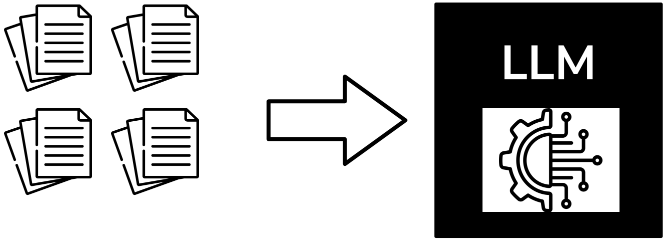
“Say something in italian”

“Reply as if you were a pirate”

“Output correct python code”



Fine-tuning



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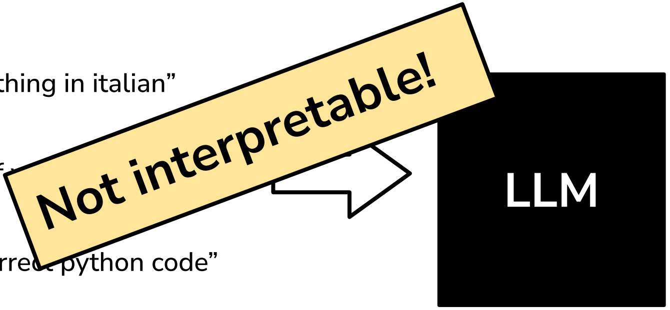

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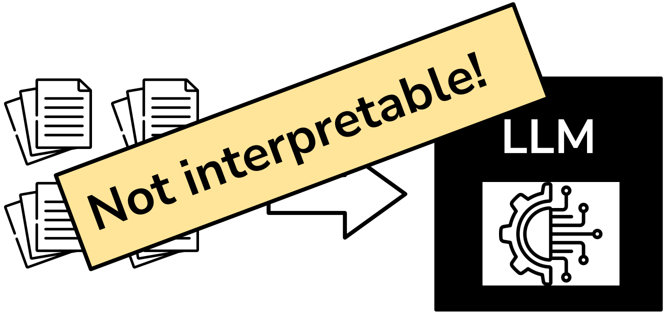
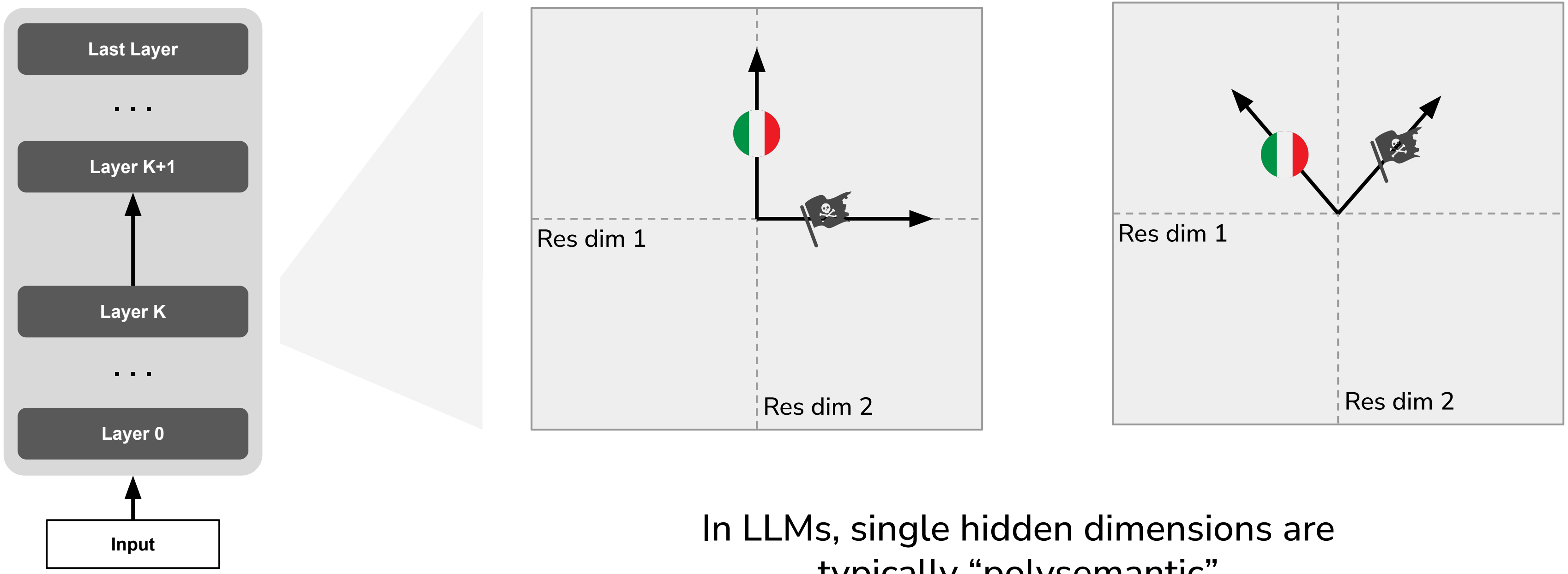


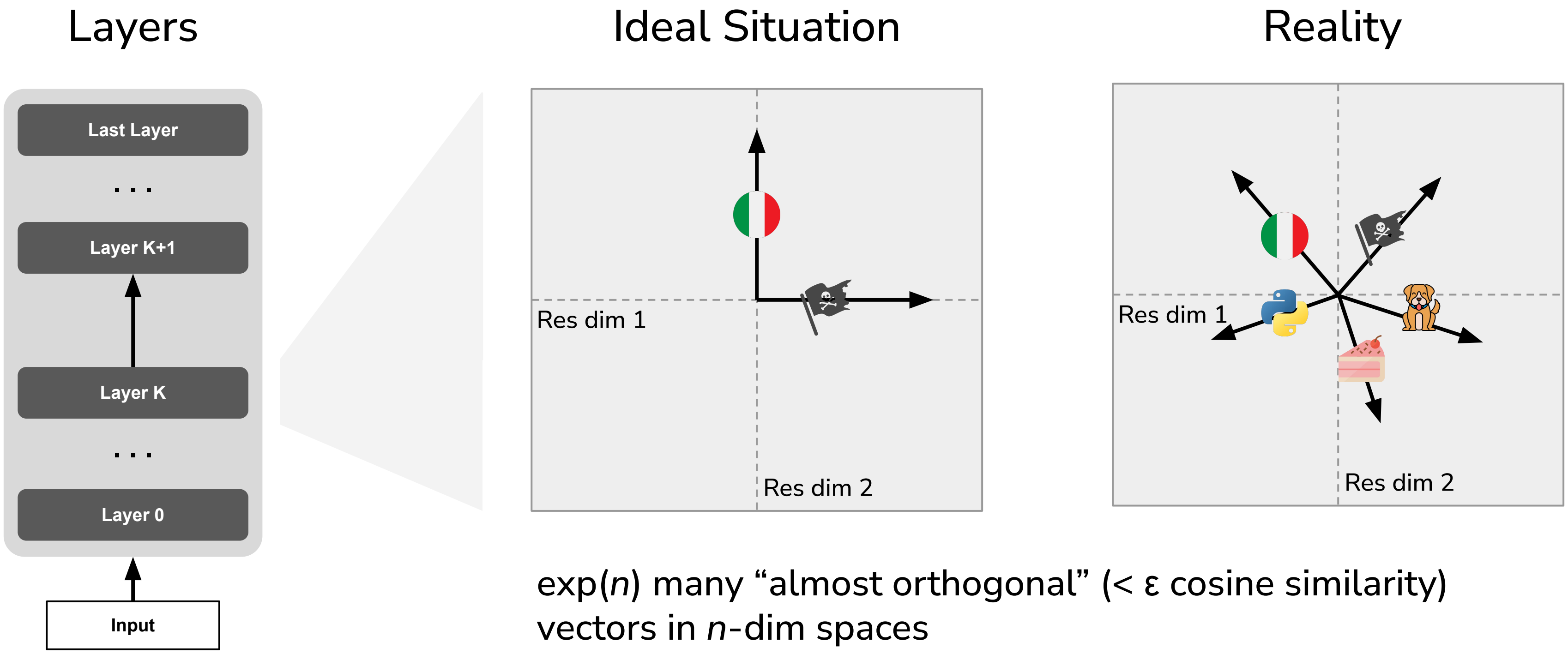
How do LLMs represent concepts?

Layers | Ideal Situation | Reality



In LLMs, single hidden dimensions are typically “polysemantic”

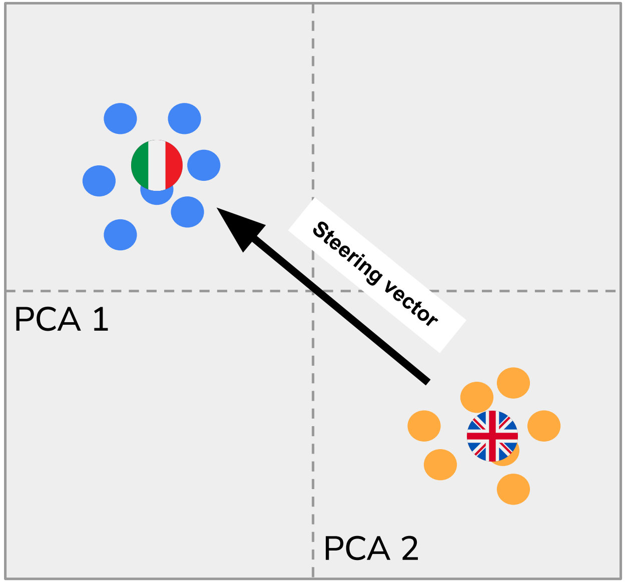
How do LLMs represent concepts?



exp(n) many “almost orthogonal” (< ε cosine similarity) vectors in n-dim spaces

Concepts could correspond to these directions!

How can we isolate directions related to concepts?



A very easy method to extract directions related to single concepts is to use a contrastive approach:

$$\mu_\ell^{\text{it}} = \frac{1}{N_{\text{it}}} \sum_{i=1}^{N_{\text{it}}} h_\ell(x_i^{\text{it}})$$

$$\mu_\ell^{\text{en}} = \frac{1}{N_{\text{en}}} \sum_{i=1}^{N_{\text{en}}} h_\ell(x_i^{\text{en}})$$

Compute means of concept-present and concept-absent

$$v_\ell^{\text{it-en}} = \mu_\ell^{\text{it}} - \mu_\ell^{\text{en}}$$

Subtract the means to get a vector in activation space that points toward the concept

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$$\hat{v}_\ell^{\text{it-en}} = \frac{v_\ell^{\text{it-en}}}{\|v_\ell^{\text{it-en}}\|_2}$$

After normalizing it, we treat it as a concept direction

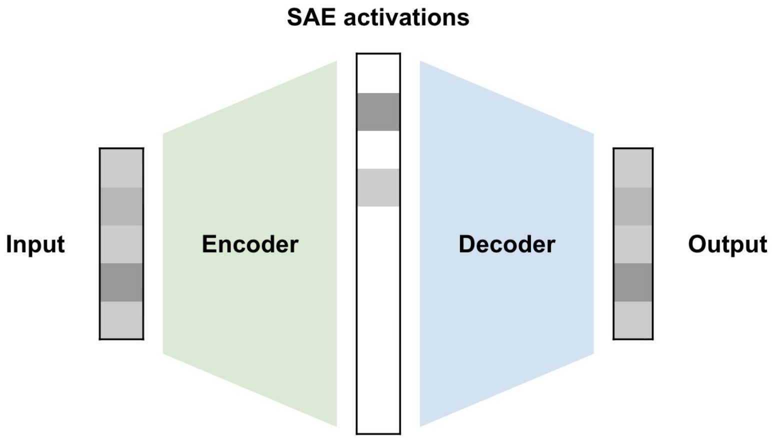
$$h'_{\ell,t} = h_{\ell,t} + \alpha\, \hat{v}_\ell^{\text{it-en}}$$

We can add the direction back to the activations to activate a concept!

Is there a more elegant way?

SAEs are trained to decompose the hidden activations of LLMs in many interpretable directions/features.

$h \in \mathbb{R}^{d_{\text{model}}}$	We start from the model activation
$u = W_{\text{enc}} h + b_{\text{enc}},$	The SAE encoder maps the activation to a higher dimensional space
$z_j = \begin{cases} u_j, & j \in \text{TopK}(u), \\ 0, & \text{otherwise.} \end{cases}$	Sparsity is ensured by retaining only the values of the top features
$\hat{h} = W_{\text{dec}} z + b_{\text{dec}} = b_{\text{dec}} + \sum_{j=1}^{d_{\text{SAE}}} z_j d_j.$	The decoder reconstructs the original activation as a **linear combination** of the decoder feature directions, weighted by the corresponding SAE feature activations.



Is there a more elegant way?

SAE features are learned without labels.
To interpret them, we can still use a contrastive approach:

$$\Delta_j = \frac{1}{N_{\text{it}}} \sum_{i=1}^{N_{\text{it}}} z_j(x_i^{\text{it}}) - \frac{1}{N_{\text{en}}} \sum_{i=1}^{N_{\text{en}}} z_j(x_i^{\text{en}})$$

Compute the means of concept-present and concept-absent values for each SAE feature, and then subtract them

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$$v_{\text{it}} = \sum_{j \in I} d_j$$

We select the SAE features associated with the bigger difference and extract a single direction summing their associated decoder directions

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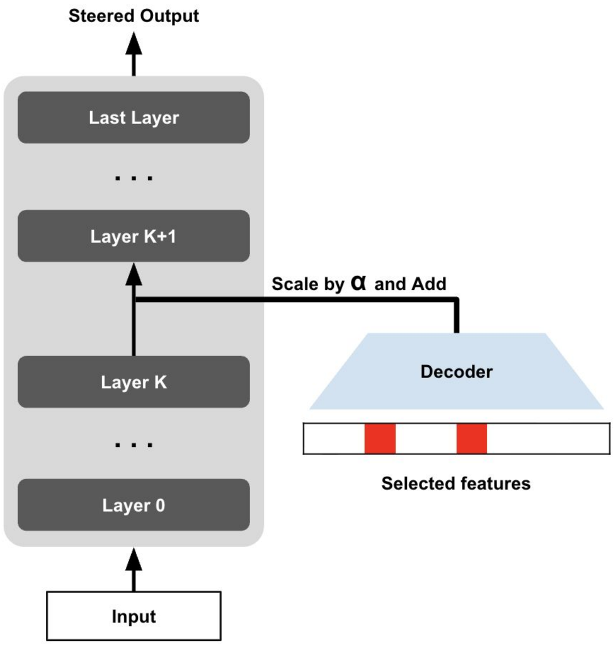
$$h'_{\ell,t} = h_{\ell,t} + \alpha v_{\text{it}}$$

We can add the direction back to the activations to activate a concept!



Now to the lab!

https://github.com/leobertolazzi/mech-interp-lab-lcl26